

Voronoi Rooms: Dynamic Visibility Modulation of Overlapping Spaces for Telepresence

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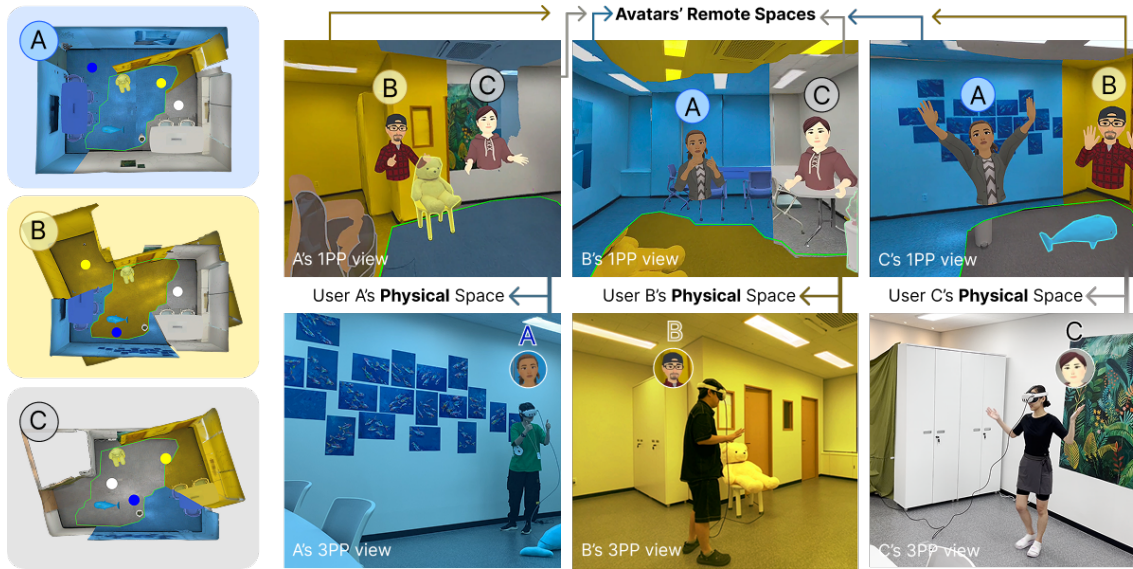


Fig. 1. The leftmost image shows the visible area from each user's standpoint (from top to bottom: Users A, B, and C's standpoint) at various positions. The positions of Users A, B, and C are indicated by blue, yellow, and white dots, respectively. The alignment of the rooms is statically determined to maximize the size of the shared space, while the visibility of individual non-shared spaces is dynamically modulated based on the users' locations. First-person (top row) and third-person (bottom row) perspective views of Users A, B, and C in three different spaces. The green line represents the shared space—an identical sub-space accessible to all users. Each user can see parts of the remote spaces around other users, augmented within their own space, which help them understand what others are doing in their respective non-shared space.

We propose a multi-user Mixed Reality (MR) telepresence system that allows users to interact by seamlessly visualizing remote environments and

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avatars overlaid onto their local physical space. Building on prior shared-space approaches, our method first aligns overlapping rooms to maximize a *shared space*—a common area containing matched real and virtual objects where all users can interact. Uniquely, our system extends beyond this shared space by visualizing non-shared spaces, the remaining part of each room, allowing users to inhabit these distinct areas. To address the issue of overlap between non-shared spaces, we dynamically adjust their visibility based on user proximity, using a Voronoi diagram to prioritize subspaces closer to each user. Visualizing the surrounding space of each user conveys spatial context, helping others interpret their behavior within their environment. Visibility is updated in real time as users move, maintaining a coherent sense of spatial awareness. Through a user study, we demonstrate that our system enhances enjoyment, spatial understanding, and presence compared to shared-space-only approaches. Quantitative results further show that our dynamic visibility modulation improves both personal space preservation and space accessibility relative to static methods. Overall, our system provides users with a seamless, dynamically connected, and shared multi-room environment. We provide an system overview and demo video of our work in the supplementary material.

CCS Concepts: • **Computing methodologies** → **Mixed / augmented reality**;

Additional Key Words and Phrases: Telepresence, space optimization, mixed realities

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1 Introduction

The motivation for this work was a vision where people’s multiple rooms are merged together as if the people are living in the same space despite their distances being far away. Telepresence, achieving the feeling of being “present together” despite the individuals being distantly located, has been a core question to address within the research community for a long time, and researchers have developed various approaches for this.

One of the early prototypes [Raskar et al. 1998] employed spatially immersive displays (SID) to surround users with room-sized displays to create an illusion of different spaces being connected. Teleport [Gibbs et al. 1999] allowed a remote virtual room to seem like an extension of the physical room. These room-level adjoining allow people to communicate as if their rooms are next to each other.

Another approach is to use augmented reality that overlays only virtual objects, such as avatars, in the real world, giving an illusion of remote users visiting the local user’s room [Jones et al. 2014]. If two spaces are dissimilar, the problem of placing an avatar to a remote space should be solved first, and for this transferring the context of the user’s interaction is considered important [Jo et al. 2015; Yang et al. 2024; Yoon et al. 2020]. To overcome spatial dissimilarity, modification to the avatar’s movement from the user’s movement to adjust to different environments becomes necessary [Choi et al. 2023; Kang et al. 2023; Wang et al. 2022]. However, visualizing only the virtual avatar may not capture the remote user’s context well especially if users interact with local objects or environment.

A different direction is to align multiple spaces to obtain a shared space, where all users can freely exist and interact with each other. Since our living space is bound by furniture and physical space limits defined by walls and floors, spatial rigidity came into consideration when securing the common ground for shared space. To overcome the spatial dissimilarity between the spaces, researchers suggested different methods to align dissimilar spaces for effective collaboration, such as changing furniture location with minimum effort [Keshavarzi et al. 2020] and partitioning spaces into multiple regions to align surfaces or partwise regions [Grønbæk et al. 2023; Kim and Woo 2023]. However, due to the differences of individual rooms, the more spaces are aligned the less shared space between multiple users will inevitably be created with the above methods.

In physical spaces, people naturally develop a shared understanding of their surroundings—furniture and other objects. This visibility of environments provide contextual cues that reveal users’ actions, interactions, and intentions. However, current telepresence systems often fail to convey this context due to limited or no visibility of

remote environments, leading to a loss of understanding of users’ intentions and movements. Moreover, current shared-space methods often leave much of each user’s environment unused. To address this, we aim to create a system where virtual replicas of remote spaces seamlessly extend from local ones, giving the illusion of organically connected living environments.

Our system is inspired by the early projection-based telepresence methods that adjoin spaces to show other users’ spaces. Early methods aimed to help participants co-exist as if they were occupying a shared physical space [Gibbs et al. 1999]. At the same time, we follow the line of work that aligns the rooms to secure a shared space for the users’ coexistence. In addition, in contrast to the previous work, our system allows users to exist not only in the shared space but also in the non-shared space, by introducing space visibility modulation. In essence, by aligning multiple rooms, a particular location outside the shared space can be overlapped by the rooms. Our key idea is to determine the visibility of the location to show each user’s behaviors as clearly as possible. Unlike previous projection-based works that only permitted static adjoining, our system dynamically changes the adjoining of spaces through space visibility modulation as the users move around freely in their rooms. From this perspective, our system fully utilizes each individual’s private space while maximizing shared space, preserving user interaction and communication. This modulation addresses the reduced shared space issue when multiple users are present, as spaces can overlap and become visible only under specific conditions—in our case, users’ positions.

Specifically, our system first aligns user rooms to maximize a shared space. This effectively overlaps the rooms so that the same area may contain elements from multiple rooms. To manage this, we apply dynamic visibility modulation, selectively displaying only the space associated with the nearest user. This is implemented using a Voronoi diagram: users act as seed points on an infinite plane, dividing the space into regions that determine which environment is visualized. As users move, the Voronoi cells update in real time, adapting visibility accordingly. Figure 1 depicts the result of dynamic visibility modulation within our system with three users.

Having a maximum shared space is important because it gives the unique impression of a remote user directly “entering” one’s space from their own as if the rooms are co-existing. At the same time, our system allows users to consistently move and show their own space to others, as our method visualizes the non-shared spaces together. As a result, users can experience a space where participating spaces are all attached together as if it is one continuous space. Our system proved effective in creating an engaging gamified experience. It also extended users’ non-shared space usage, fostered a sense of togetherness, and increased interest toward other spaces.

Our system offers benefits in several scenarios, such as space-based collaboration like architecture or interior design meetings, VR training, education, games, and social events. In addition, space adjoining can address social isolation issues by creating an illusion of “living together” [Maimone and Fuchs 2011b].

With this work, our research contributions are as follows:

- We introduce a novel concept that dynamically adjusts the visibility of multiple physical spaces based on users’ positions in a multi-user telepresence environment. Users can fully

utilize their own space, as the same area can contain multiple space information through visibility modulation.

- We developed a fully workable MR system that allows users to communicate with each other using their physical spaces.
- We conducted a user study to validate the effectiveness of our system.

2 Related Work

Researchers have studied telepresence with a desire to bring remote people into local space. One of the early works was proposed by Raskar et al. [1998], a proof of concept of “spatially immersive display” surfaces using multiple cameras. Through these dynamic surfaces, distant individuals and their spaces are reconstructed and projected to the local surfaces. The result is a “through-the-window” paradigm, creating the illusion that collaborators are located side-by-side despite being physically distant. Another pioneering example is Cave Automated Virtual Environment (CAVE) [Cruz-Neira et al. 1993], which serves as a general-purpose SID. These projection-based solutions allow multiple viewers to engage simultaneously within the system.

Head-Mounted Displays (HMDs) combined with 3D capture and reconstruction technologies have revolutionized telepresence by enabling the transmission and display of 3D replicas of remote objects or people within local spaces [Jiang et al. 2024; Matsuki et al. 2024; Xu et al. 2023]. With the continuous interest toward telepresence within the graphics community and the continuous refinement and improvement of graphics technology related to HMD, the development of various methods advances the telepresence research further [Lawrence et al. 2022]. This advancement contributes by significantly enhancing the sense of copresence for local users.

Maimone et al. [2011a] introduced a telepresence system with real-time 3D scene capture and continuous-viewpoint head-tracked stereo 3D display. They further proposed room-level reconstruction and transfer to the other user’s space using displays, creating the illusion of rooms facing each other [Maimone and Fuchs 2011b]. Their subsequent work [Maimone et al. 2013] incorporated an optical see-through head-worn display, allowing remote users to be observed from the correct viewpoint without unnatural occlusion. Beck et al. [2013] developed a system enabling distant groups to meet in a shared 3D virtual environment projected onto a plane, using depth cameras to continuously capture and transmit multiple users’ locations. Pejsa et al. [2016] advanced this concept by transcending a 3D capture of the local user to the remote space, creating an illusion of physical face-to-face conversation. The field advanced again with Holoportation, which achieved high-quality capture and transmission of remote people and objects for telepresence [Orts-Escolano et al. 2016]. In more recent work, Lawrence et al. [2021] introduced a telepresence system that does not require HMD or body-worn gadgets through image-based geometry fusion to achieve the transmission of 3D bi-directional communication.

To address the challenges posed by the dissimilarity between telepresence environments, researchers have focused on placing and animating avatars to preserve the semantics of the remote user’s movement. Various approaches have been developed to determine optimal avatar locations. Jo et al. [2015] proposed using spatial object-level characteristics between remote and

local sites to guide avatar placement. Yoon et al. [2020] advanced this concept by employing neural networks to learn and predict suitable spatial relationships. More recently, Yang et al. [2024] introduced visual guidance systems to suggest avatar positions based on users’ interaction targets. These instant transportation of avatars are fast to achieve and preserve spatial relationships between the remote and local. Other works attempted to modify avatar motion: the locomotion needs to be adjusted when local and remote spaces have similar but different scale or furniture layout [Choi et al. 2023; Kim et al. 2016; Wang et al. 2022], and the avatar’s deictic gesture needs to be modified if pointing targets get varied [Kang et al. 2023].

Various telepresence approaches are possible if we allow merging, overlapping, scaling, or dividing spaces or objects to support interaction between users. These include aligning areas with similar features from two users’ physical spaces [Lehment et al. 2014], or reconstructing and displaying portions of the remote space [Thoravi Kumaravel et al. 2019]. For multiple users in asymmetric environments, access to the same scanned environment can be provided [Irlitti et al. 2023]. Additionally, using augmented cues to highlight each user’s spatial awareness can increase social presence [Fink et al. 2022]. Adjustments to scale and deictic orientation can be another direction to accommodate HMD’s limited field of view [Piumsomboon et al. 2018]. Removal of HMD itself from the collaborator’s viewpoint can be another solution as Frueh et al. [2017] suggested. Their method erases the HMD from the view of the collaborator and transmits synthesized gaze based on machine learning and graphics techniques.

Understanding scene layouts and semantics benefits creating effective telepresence spaces. Moving furniture of multiple users to create maximum mutual space while minimizing physical effort [Keshavarzi et al. 2020], finding mutual functional space [Keshavarzi et al. 2022] has been proposed. Partial alignment through partitioning space with respect to flat task space’s edge and then rescaling the virtual space [Kim and Woo 2023], or the interaction methods analyzing users’ preference toward different partial alignment method [Grønbaek et al. 2023] has been explored. Kim et al. [2024b] proposed partial alignment from the object cluster registration aspect, which identifies object-wise relationships between different spaces to align regions. Recently, SpaceBlender [Numan et al. 2024] suggested a merged VR environment using images of multiple users’ physical spaces into one so that contextual cues can be transmitted properly. While their work focused on generating a virtual environment, our work focus on utilizing users’ physical space fully.

Aiming for collaborative VR, researchers have conducted studies to overcome users’ spatial differences through mapping or fusion techniques as well. Congdon et al. [2018] optimized spaces to maximally preserve dynamic walking trajectories from local to remote environments. Sra et al. [2018] introduced three physical-virtual mapping models: scale, kernel, and overlap, for aligning physical rooms with diverse sizes to the shared virtual room space to enjoy contents together. Fused scenes can encourage fast task achievement as Wang et al. [2023] showed by a VR scene-stitching. Using a predefined straight line as a region to fuse, and the Bezier curve as a fusion parameter, they stitch several rooms into one virtual environment to increase other users’ room environments.

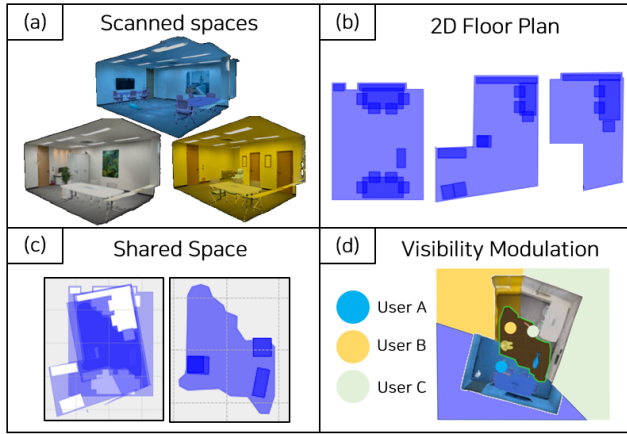


Fig. 2. Our system modulates visibility based on user positions, allowing multiple users to utilize each space fully, with the following procedure. (a) We first scan the physical spaces to obtain 3D models. (b) 2D floorplans are extracted from 3D models. (c) The spaces undergo an optimization process to maximize shared empty space for all participating spaces, followed by object augmentation to maximize the shared space. (d) In the MR system, non-shared spaces are split in real-time based on Voronoi regions for visibility modulation.

Redirected walking techniques overcome physical space limitations to allow users to navigate in larger virtual spaces [Razzaque et al. 2001]. Building on this principle, Sun et al. [2018] redirected users during temporal blindness to avoid static and dynamic obstacles in room-scale VR, and Langbehn et al. [2018] similarly exploited temporal blindness to imperceptibly adjust users' position and orientation. Addressing multi-user settings, Dong et al. [2019] proposed a redirected smooth mapping that subtly distorts the virtual scene to guide users, and Jeon et al. [2022] further advanced this direction by using reinforcement learning to dynamically partition shared space and minimize discomfort, such as excessive resets. Taken together, these fusion-based and redirected walking approaches share a common goal: maximizing the effective use of physical space while minimizing disruptions to natural user movement.

Recognizing that people inhabit diverse rooms and houses, we follow the view that leveraging space alignment is necessary. At the same time, we aim to increase the space usability through overlapping and visibility modulation. Our work enhances the sense of presence by creating an environment where not only the other person but their entire living space is brought together.

3 Overview

Figure 2 illustrates the workflow of our system. Our system requires scanned 3D model of the physical spaces, from which a 2D floorplan of each space is obtained as a polygonal mesh. Our method first maximizes mutually shared space where all users can move around without any obstacles through overlapping room alignment. The alignment procedure consists of three phases: empty space alignment, object augmentation, and boundary smoothing. From 2D floorplan, our method extracts all the collision-free space where each user can move freely. Then, the alignment of the rooms is

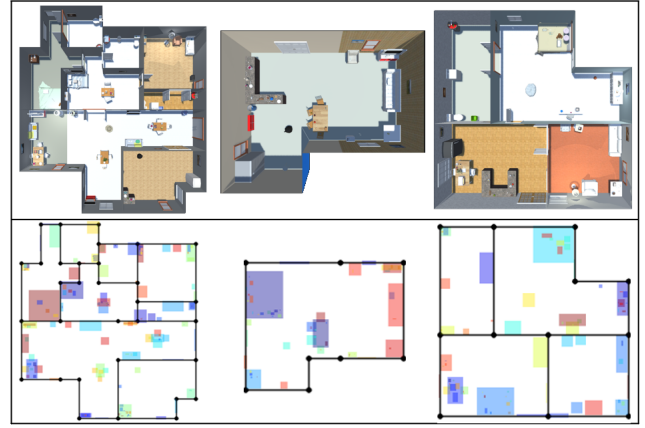


Fig. 3. House models used for the study. The first row is the top view of the original house model while the second row is the 2D polygonized floorplans.

determined such that the overlapping region of the empty spaces is maximized, which secures a maximal space for users to interact with each other. The subsequent object augmentation phase augments virtual objects to make the shared space identical within each room. Finally, the boundary smoothing phase removes sharp corner areas that cannot be practically used for a shared space.

Given the space alignment, we determine the visibility of the non-shared spaces based on the user's locations. Specifically, the visibility of the non-shared spaces is determined to maximize the area of non-shared space around each user's current position using the Voronoi split. This strategy is to convey the context of the remote user's movement in relation to their surrounding space clearly. As a result, part of remote users' non-shared spaces are augmented in the local space around the remote user's avatar, giving an illusion that the remote space is attached or connected to the local physical space. Our dynamic visibility modulation scheme allows users to move freely within their rooms, regardless of whether the location is within the shared space or not. With the same logic, users can observe the full spaces of other users as well.

3.1 Room Data Representation and Processing

We used ProcTHOR [Deitke et al. 2022] as a dataset for VR demonstration. Given a 3D house as a 3D mesh, we construct a 2D floorplan by projecting the room component onto the horizontal plane as depicted in Figure 3. For floors and walls, we extract the boundary edges of plane mesh as floor plan lines. For objects, we construct an oriented bounding box for each object. We suppose there are n number of spaces S_i ($i = 1 \dots n$) to be merged, and for simplicity each space is occupied by one user.

4 Method

This section details our space alignment and dynamic visibility modulation procedure. The input to our system is multiple spaces represented as a 2D floor plan. We select a single room within a full house from the ProcTHOR dataset, including floors, walls, and objects for space optimization. For the MR system, we use a scanned dataset reprocessed into a 2D floor plan.

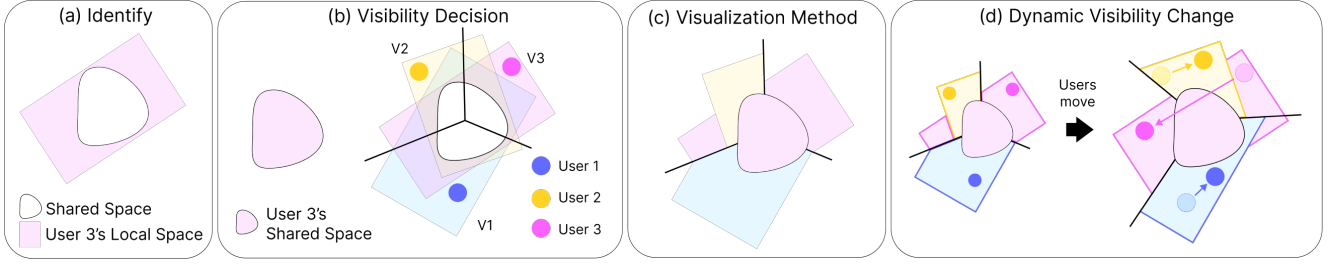


Fig. 4. This diagram shows the visualization process with the Voronoi partition. (a) The boundary between shared and non-shared space is computed and input into the system. (b) Inside the shared space, each user's local physical space is directly visible through the head-mounted display's pass through feature. For the non-shared space, the Voronoi diagram determines which areas of the rooms to visualize. (c) For regions outside the shared space, pixels are rendered according to users' corresponding Voronoi regions. The remaining spaces; the shared space and the local space that are not overlaid and rendered by other spaces, are shown as pass through. (d) As users move, their positions update the Voronoi diagram, dynamically adjusting visible areas to reflect changes.

4.1 Optimal Alignment of Rooms

4.1.1 Maximizing Shared Empty Space. The first part of our optimization procedure follows a similar procedure of finding maximized empty spaces through spatial alignment with previous works [Keshavarzi et al. 2020, 2022; Lehment et al. 2014]. Our optimization starts with extracting an empty area. Empty area refers to the volume of a space where no objects fall within a human user's height range. In these areas, users can move freely without the risk of colliding with any objects in their physical space. We calculate this area by extracting all objects from the floor plan as Equation (1)

$$E_i = S_i - \bigcup_{k=1}^{m_i} O_{i,k}, \quad (1)$$

where S_i is each user i 's space, $O_{i,k}$ is the object k 's area, and m is the number of objects within each space.

The alignment of the rooms is conducted to determine the position p_i and rotation θ_i of the rooms to maximize the area of overlapping empty spaces.

$$A_e = \max_{p_i, \theta_i} \bigcap_{i=1}^n E_i. \quad (2)$$

The intersection A_e of the aligned empty spaces is identified as the shared empty space. We use the simulated annealing method for optimization, with the initial alignment set so that the centroids of E 's coincide.

4.1.2 Object Augmentation. Adding on previous approaches, our method introduces an object augmentation phase as the second component of our optimization framework. In the object augmentation phase, objects are added to the shared empty space A_e by incorporating digital copies of physical objects from one space into the other spaces. Augmenting objects expands the shared space, allowing interaction between users and the augmented objects. For example, consider a region within multiple spaces where only one space contains a chair. Without object augmentation, this region would not be considered shared space since other spaces lack corresponding chairs. However, by augmenting the chair, the region it occupies becomes accessible to all participants. We augment only those objects located within the boundary of A_e , as our goal is to create a continuous, unified shared space for all participants. In addition, we reject an object from augmentation if it overlaps with

other objects in participating spaces to avoid confusion.

$$A = A_e \bigcup_{i,k} \text{Augmentable}(O_{i,k}), \quad (3)$$

where the function *Augmentable* checks whether an object $O_{i,k}$ satisfies the boundary and non-overlapping conditions.

4.1.3 Shared Space Polygon Trimming. The raw output of the maximized shared space includes narrow areas and regions that are practically inaccessible to people. Therefore, we use a flood-fill approach to smoothly retract the outline of the shared space inward.

4.2 Dynamic Visibility Modulation

The room alignment process causes the participating spaces to overlap. Within the shared space, users can interact as if they were in the same physical environment. However, seamless interaction is not possible in the non-shared space, as users cannot coexist there. This is due to differences in overlapping regions between spaces. For instance, the same region may contain furniture in one space but not in others, or a region inside one room may fall outside the boundary of another.

To address this issue, we allow users to remain in their own non-shared spaces through dynamic visibility modulation, which selectively visualizes only one of the overlapping spaces. The selection is based on showing user actions as clearly as possible, displaying only the space associated with the nearest user. This approach enables users to see what others are doing in the non-shared spaces, even though they cannot enter another user's non-shared space.

As Figure 4 shows, the dynamic visibility modulation can be achieved straightforwardly using a Voronoi diagram, where points scattered on a plane subdivide it into cells, with each cell enclosing the region closest to a given point. In our case, we assume an infinite plane with points representing each user. The Voronoi diagram divides the plane into cells, with each cell corresponding to the region where user i 's non-shared space will be visualized.

$$\text{Cell}(i) = \{x \in \mathbb{R}^2 : \|x - p(i)\| \leq \|x - p(j)\|, \forall j \neq i\}, \quad (4)$$

where x is any point within a plane excluding the shared space and $p(i)$ denotes the position of user i . As users move around the room, the Voronoi diagram is dynamically calculated in a shader program for Algorithm 1 to reflect their movements.

ALGORITHM 1: Visualization of Remote Space for User i

```

1: for each Space  $j \neq i$  do
2:   for each fragment  $f$  of Space  $j$  do
3:     if  $f \notin A \wedge f \in \text{Cell}(j)$  then
4:       Draw  $f$ 
5:     else
6:       Discard  $f$ 
7:     end if
8:   end for
9: end for

```

4.2.1 Visualization of the Spaces. In our MR system, each user's local space is visualized through video pass through, while the non-shared spaces of other users are displayed as 3D digital replicas, occluding parts of the local user's physical space. As a result, each user perceives the shared space as part of their local environment, creating the illusion that other users are physically present in their space.

The visualization of the Voronoi partition of other spaces can be efficiently performed in real-time using shader computation. We implement the visualization by determining which fragment within each object to show or not. For rendering to User i 's display, we determine whether to draw a fragment f of other spaces or not by Algorithm 1.

The Voronoi partition remains consistent across both shared and non-shared spaces. This is because Voronoi regions are calculated using users' world positions synchronized through the server. The fragment rendering of all spaces occurs simultaneously, without any sequential ordering between different spaces. Each shader fragment is rendered directly based on the Voronoi region it belongs to, determining the final display.

5 Results

We conduct several experiments to demonstrate how our method performs quantitatively across different spaces. The first evaluates how our dynamic algorithm performs with diverse spatial mixtures. The second compares our dynamic visibility modulation over static visibility scheme.

We constructed a test dataset by randomly selecting 150 houses containing 511 rooms from ProcTHOR dataset [Deitke et al. 2022]. The mean and standard deviation of the room sizes are $26.0m^2$ and $16.4m^2$, respectively. To examine how the number of participating rooms affects the results, we randomly sampled rooms from the dataset and created 50 combinations for each room count.

5.1 Quality of Shared Space

Figure 5(a) shows the relationship between the number of spaces and the resulting size of the shared space. As the number of spaces increases, the overall size of shared areas progressively decreases. Assuming 90 cm as an acceptable interpersonal distance [Sorokowska et al. 2017] and the users are placed in a grid pattern, a rough calculation tells that on average a shared space made by more than 8 spaces cannot accommodate all users at the same time. However, notably our system allows users to communicate with each other even when they are in a non-shared space.

Figure 5(b) shows that the number of augmented objects decreases as the number of spaces increases. With only two spaces,

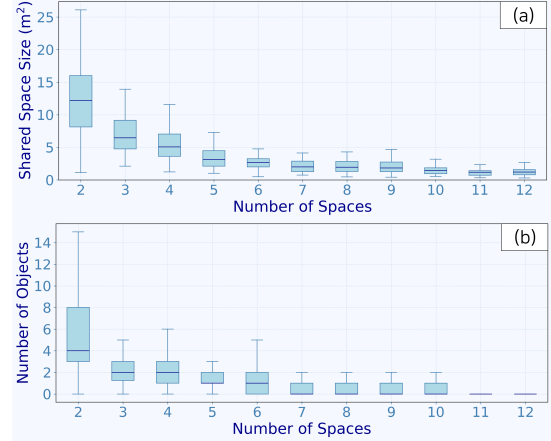


Fig. 5. Shared space quality as the number of spaces increases. (a) Shared space size. (b) Number of augmented objects in shared spaces.

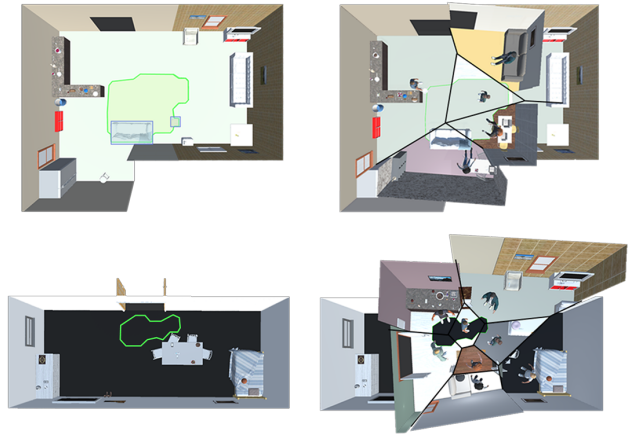


Fig. 6. Example of five (top) and eight (bottom) overlapping rooms participating in telepresence, as seen from a room on the left. The shared space and augmented objects are visualized with a green line and bounding boxes.

the number of augmented objects varies as augmentation depends largely on their relative sizes; if one fully contains the other, all objects in the smaller space are augmented. However, as more spaces are added, the available shared areas shrink, reducing the potential for object augmentation, as fewer objects occupy positions intersecting with the shared space.

Figure 6 presents visual results of our method when five and eight rooms are accommodated. As shown in Figure 5, the spaces look more cluttered as the number of rooms increases. Still, users are not constrained within the shared space and can keep interacting with others in the non-shared spaces.

5.2 Comparison between Static and Dynamic Visibility Modulation

We evaluate the advantages of our dynamic visibility modulation method (Dynamic) over a static visibility scheme (Static). In the

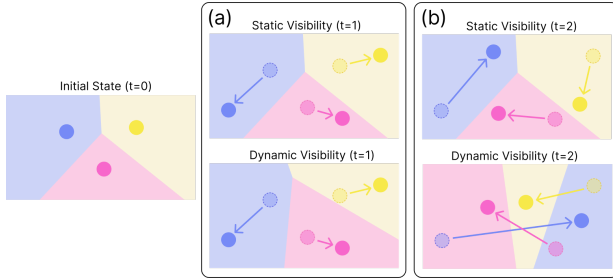


Fig. 7. Comparison of static and dynamic visibility over time. (Left) Initial state ($t=0$) with three users (blue, yellow, pink) and their Voronoi regions. (a) At $t=1$, when users move to the same positions, the visibility is updated for the dynamic scheme while it is fixed for the static scheme. (b) In the static environment, users are restricted to initial regions, while in the dynamic environment, they can move freely within their local spaces.

Static method, the visible area for each user is fixed after the initial Voronoi partitioning based on their starting positions, regardless of subsequent movement (Figure 7). As a result, user movement is constrained to their initial visible region.

To compare the Static and Dynamic methods, we use a random sampling of user positions. For Static, the sampling process occurs in two phases. In the initialization phase, positions are sampled only within shared spaces to ensure connectivity between shared and non-shared areas—this choice favors the Static method by maximizing each user’s navigable area. From the second iteration onward, sampling includes both the user’s Voronoi region and the shared space, simulating realistic movement within assigned areas. In contrast, the Dynamic method allows users to move freely within their rooms at all times, with positions sampled randomly throughout.

5.2.1 Personal Space Preservation Ratio. First, we evaluate how effectively personal space is preserved as the number of overlapping rooms increases. We define “personal space” as the area surrounding a user that should remain visible to others to convey spatial context. Based on interpersonal distance studies [Sorokowska et al. 2017], we model personal space as a circular region with a 90 cm radius. For each randomly sampled user position, we measure the proportion of this personal space that lies within the user’s visible region. For example, if a user’s entire personal space falls within their boundary, it is considered 100% preservation.

Figure 8 (top) shows a natural trend that the rate of personal space preservation decreases as the number of participating rooms increases. Comparing the Dynamic and Static methods, the Dynamic approach consistently preserves personal space more effectively—maintaining 60% preservation even with 12 overlapping rooms. In contrast, the Static method exhibits a sharp decline in preservation, as users near Voronoi boundaries suffer from significant visibility loss.

5.2.2 Accessible Area for Each User. We analyze the accessible area available to each user under both methods. In the Static approach, a user’s accessible area is restricted to the shared region and the fixed Voronoi region assigned to their space. As the number of spaces increases, this constraint leads to a reduction in accessible area. In contrast, the Dynamic method allows users to access their entire room, consistently achieving 100% accessibility, as shown in

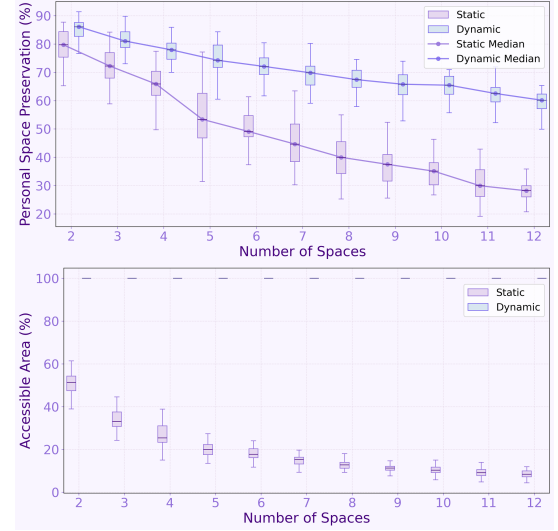


Fig. 8. Relationship between number of spaces and spatial qualities in static versus dynamic visibility. (Top) Average ratio of personal space preservation. (Bottom) Average ratio of accessible area over entire space.

Figure 8. This perfect accessibility score highlights the Dynamic method’s ability to fully accommodate each user’s spatial context and environment without imposing fixed spatial boundaries.

6 Mixed Reality User Evaluation

To understand the user experience of our telepresence system, we conducted a user study where 33 participants performed two tasks in an environment shown in Figure 1. We make the following research questions that mainly focus on how users would accept the generated space where multiple rooms are visualized.

- Given that all users can see the full area of others’ non-shared space by moving from place to place, we expect users to have more understanding of other rooms while using each of their physical spaces more. We assess this through users’ trajectories and questionnaires.
- Our visualization would enhance spatial presence since remote spaces would seem to be stitched to the local user’s space. We evaluate this using multiple questionnaires and semi-structured interviews.

6.1 Study Conditions and Measures

6.1.1 Conditions. As shown in Figure 9, we compare two conditions, Baseline and Ours, in our experiment as follows:

Baseline: Without non-shared space visibility. Serving as a baseline to compare, Baseline uses only shared space within each user’s space. Creating mutually shared space is widely used in multi-user telepresence environments such as Keshavarzi et al. [2022]. Users only saw the shared space line and their own physical space. In this condition, users cannot see the other users’ spaces but only their own. Only the objects augmented within the shared space were visualized to all users. Furthermore, the other users’ avatars would disappear if they step away from the shared space.

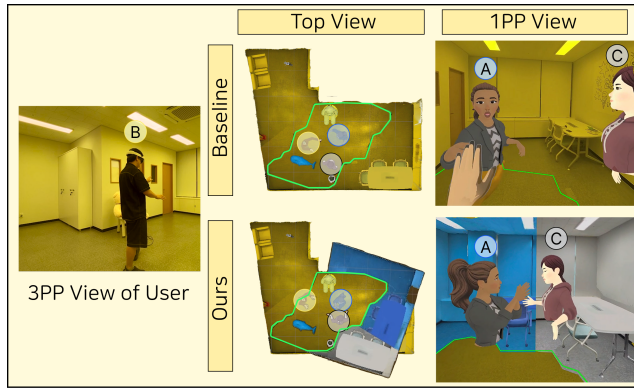


Fig. 9. Visibility in the Baseline and Ours conditions from similar fields of view. The left column shows a third-person view of User B. The center column presents top-down views: in Baseline, only the local space is visible, while in Ours, remote spaces are visible depending on the Voronoi region split. The right column shows the first-person perspective (1PP view) of User B.

Ours: With non-shared space visibility. Ours represents our suggested method. Users saw the shared space line, their own space, and other users’ spaces visualized depending on users positions.

With two conditions, we construct the following hypothesis.

- H1: In Ours, users will report an increased understanding of the other users’ spaces.
- H2: The message understanding would be higher in Ours since the avatars can always be visible to each other.
- H3: In Ours, users will report higher social presence as not only avatars but their spaces are visualized together.
- H4: In both conditions, users will report a relatively high sense of presence toward the presented space.
- H5: In both conditions, users would feel interested.
- H6: In both conditions, the presented space would not cause much inconvenience to users.
- H7: In Ours, users will make more use of their own space. Their usage extends from just using the shared space to spreading to other personal spaces which will be shown through increased trajectory area and amount.

6.1.2 Measures. To verify our hypothesis, we collect both subjective and objective measures. For the subjective measures, we derive questionnaires from previous studies. To verify users’ experience with generated space, we use the **MEC spatial presence questionnaire (MECSPQ)** [Vorderer et al. 2004]. Specifically, Spatial Situation Model (hereafter referred to as ‘MySpace’ and ‘YourSpace’) and Spatial Presence were used to understand users’ comprehension of remote spaces and their presence. For MySpace, YourSpace and Spatial Presence, questionnaire included 4 items rated on a 5-point Likert scale (1: I do not agree at all to 5: I fully agree). We wanted to understand whether our dynamic system would cause any discomfort so we included **Virtual Reality Sickness Questionnaire (VRSQ)** [Kim et al. 2018]. VRSQ included 9 items rated on a 4-point Likert scale (0: not at all, 1: slightly, 2: moderately, and 3: very). We used the Intrinsic Motivation Inventory [Edward McAuley and Tammen 1989], especially questions related to interest/enjoyment

to access the users’ level of interest (‘Interest’). Interest questionnaire included 7 items rated on a 7-point Likert scale (1: never to 7: very true). We measure social presence using a questionnaire from Temple Presence Inventory (‘TPI’) [Lombard et al. 2009] to see whether having stitched spaces would increase social presence. TPI questionnaire included 7 items rated on a 7-point Likert scale (1: never/not at all to 7: always/very much). Also, to analyze movement specifically, we adopted questions from Harms and Biocca [2004] (‘Message Understanding’) with modifications to focus more on movement. We brought 4 items from the original Message Understanding questionnaire. Then, we modify the word “thoughts” to “movements”. Participants responded on a 7-point Likert scale ranging from 1: I do not agree at all to 7: I fully agree. For objective measures, we collected the full trajectory of the user on each condition. The method to track the user’s trajectory is written in detail in the system implementation.

6.2 Task

The study task was chosen based on the following criteria: (i) it requires collaboration between users, (ii) it encourages user movement within their space, (iii) it relates to real physical space, (iv) it is not overly demanding, and (v) it promotes space searching and understanding. Along with these criteria, and aiming to analyze how our space visibility system affects the user’s MR spatial experience, we designed two tasks: (1) Introducing Space Task and (2) Block Stacking Task, to do within one condition as illustrated in Figure 10. The first task involves introducing objects within their space, serving as an icebreaker, and encouraging free conversation between participants.

The second task is a puzzle involving collaborative block stacking with a specific order. We chose this since block matching using Lego or puzzles has been widely used for local-remote collaborative tasks [Bataille et al. 2023; Tian et al. 2023; Wang et al. 2021]. Block visibility varies based on the conditions. In Baseline, blocks outside the shared space are only visible to the space owner. To show a block to other users, it must be brought into the shared space. In Ours, users’ locations influence the visibility of other spaces and their blocks. Blocks inside the shared space are always visible to all users. However, blocks outside the shared space may not be visible, even to the space owner, as they can be occluded by other visualized spaces. In both conditions, blocks outside the shared space can only be moved by the room owner. We assign different colored blocks to each user, and users are restricted to grabbing blocks of their assigned color. This is to ensure that all three participants could contribute equally, rather than having one person stack all the blocks.

6.3 Experiment Space Construction and Scanning

We conducted experiments with groups of three participants at a time. This choice aligns with our research goal of demonstrating a multi-user telepresence system through visibility modulation, using all space fully. This three-person implementation effectively demonstrates the system’s scalability while maintaining experimental control.

For our experimental purposes, participants needed to quickly grasp each space’s unique character, as their task involved introducing their spaces to one another. To facilitate this, we designed three separate spaces with different interior styles (aquarium, pop



Fig. 10. User C's first-person perspective (1PP) views from white room are shown for each combination. In Baseline (top row), non-shared spaces are not visible. In Ours (bottom row), non-shared spaces are visible. The left column displays screenshots of Task 1, introducing spaces. The right column shows users performing Task 2, block stacking.

art, and nature styles), and different lighting for participants to easily distinguish the spaces as shown in Figures 1 and 2. We chose lighting color of blue, yellow, and white as they are easily distinguishable while not bringing negative emotions as red color [Elliot and Maier 2014; Kaya and Epps 2004] or being discolored through HMD display as green color. Yet, it is to be noted that our method is not bound to be used only for different spaces. Rather, indistinguishable spaces might be ideal for a seamless integration between spaces. Each room as in Figure 1 (a), (b), and (c) has a size of $33.47m^2$, $29.81m^2$, and $21.08m^2$ respectively with all of its objects included. All rooms have the same height.

After constructing the three spaces, we scanned the rooms using existing software [LaanConsultingCorp 2024; Polycam 2024] to obtain their 3D models. In addition, we reconstructed the furniture 3D models with Luma AI software [LumaAI 2024].

6.4 MR System Implementation

We selected the Meta Quest 3 as the HMD device for our experiment due to its video-see-through capability. For system development, we used the Unity game engine (2022.3.13f1) along with the Meta XR Core SDK. We created the avatars using the Meta Avatar SDK, and implemented multiplayer networking with Unity Netcode. For rendering and system control for the three-person user demo, we used two desktops and one laptop, each with a minimum CPU of Intel Core i7 and a Titan XP graphics card. Within our system, each user's visibility of their own space and the other space should be different and dynamically modulated. For example, the user should be able to observe their own space within the shared space using the see-through function, and observe other users' non-shared spaces outside the shared space.

6.5 Procedure and Participants

In our system, three users participated at a time, each in a separate physical room. Upon arrival, participants provided demographic

information and signed a consent form. We then explained the experiment, including the number of conditions, approximate duration, and tasks for about ten minutes. Participants were assisted with wearing the HMD, and once they were ready, the session began.

The session started by adjusting the see-through function of the HMD headset. Users walked around their respective physical rooms for one minute. After adjustments, each condition began. When all three users entered the system which took two to three minutes, we started the first part of our experiment: introducing the rooms. Each user moved around their room and described its interior and structure, with two minutes allocated per participant. Depending on the condition, other rooms might or might not be visible. After the room introductions, and once users were familiar with the system, we moved on to the second part of the experiment: the block stacking task. This phase lasted four minutes. Blocks were distributed across all three rooms, with each user assigned three blocks—one inside the shared space and two in their non-shared space. The visibility of the blocks varied depending on the conditions. Participants had to collaborate to compare the sizes of the blocks and complete the tower.

Participants experienced two conditions in total, with the order randomized for each participant to minimize potential ordering effects and to remove the fatigue effect that might happen for the second trial. After each condition, participants completed a questionnaire, which took five minutes on average to complete despite a slightly long questionnaire, similar between both conditions. As such, participants spent approximately 21 minutes in each condition, including the time for instruction given at the start of each task. None of the participants asked to stop the experiment due to discomfort. Once all conditions were finished, we conducted semi-structured interviews for about seven to eight minutes to gain in-depth insights into their preferences and opinions. The entire experiment took approximately 60 minutes per participant.

Our experiment was reviewed by the Institutional Review Board. We recruited 33 participants (15 male, 18 female; age 22–35, $M=25.96$, $SD=2.57$) through campus announcements and obtained written informed consent. Participants were compensated for their participation after the experiment. For VR experience, 4 participants had never used VR, 22 used it rarely (a few times a year), 4 used it regularly (at least once a month), and 3 used it frequently (at least once a week). For MR experience, 21 had never used it, 10 used it rarely, 2 used it regularly, and none used it frequently.

6.6 User Evaluation Results

6.6.1 Results of Objective Measure. We analyzed the participants' trajectories as shown in Figure 11 (a). We extracted trajectory only from Task 1 (Introduction Task) as the location of blocks influenced trajectories in Task 2. Our analysis focused on two key metrics: (i) the average distance from a reference position and (ii) the maximum deviation from a reference position, for each experimental condition. These metrics quantify how user trajectories differed between conditions.

We calculated the reference point of each user's trajectory as the mean position of all points in their path. We recorded the user's position 0.1 seconds to track the trajectory. For the first metric, we calculated the mean distance from the centroid by

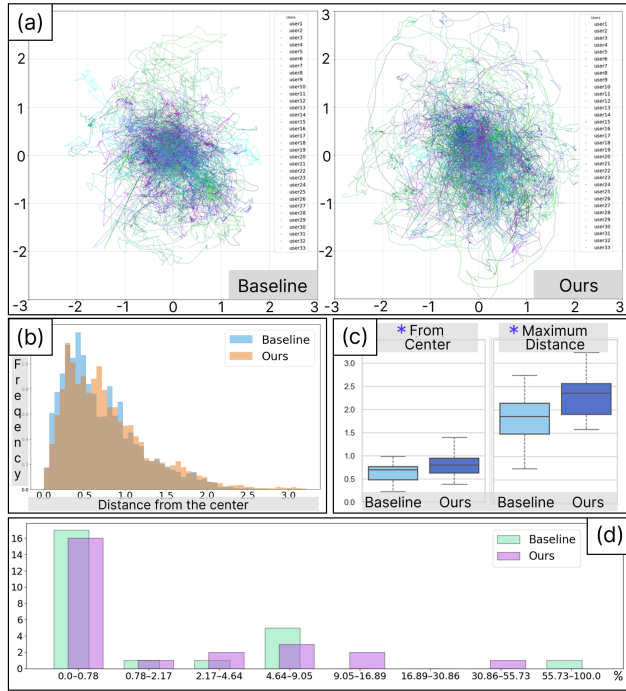


Fig. 11. Overview of trajectory analysis. (a) All user trajectories from user experiments between Baseline and Ours. The unit is meters. (b) Distribution of user distances from the reference position. (c) Statistical result of trajectories. Both measures show significance. (d) Spatial distribution obtained by dividing the area in (a) into a 5×5 grid and plotting the number of grids by the percentage of trajectory points (log scale).

summing the distances of all points traversed by a user during the task and dividing by the total number of recorded positions. This average provides insight into the spread of participants' trajectories. For the second metric, we identified the maximum distance from the centroid for each user in both conditions, capturing the extremity of the trajectories. This dual approach examines both the overall spread of user movement (through average distances) and the range of movement (through maximum distances), providing a more comprehensive understanding of how different conditions might influence user behavior in the virtual environment.

For both (i) and (ii), the data did not pass the normality test using the Shapiro–Wilk test [Shapiro and Wilk 1965]. Thus, we report the result of the Wilcoxon signed-rank test [Conover 1999; Wilcoxon 1945] for the analysis as shown in Figure 11(c). For (i), the difference between conditions Baseline and Ours showed significance ($p = 0.018$, $W = 136.500$, $z = -2.384$). Condition Ours showed a larger average distance ($m = 0.780$, $sd = 0.213$) than the Baseline ($m = 0.678$, $sd = 0.269$), about 15% higher. For (ii) as well, the difference between conditions was significant ($p < 0.001$) with the statistical value of $W = 89.000$, $z = -3.422$. The maximum distance was higher in Ours ($m = 2.300$, $sd = 0.446$) than Baseline ($m = 1.807$, $sd = 0.503$). As a supplementary analysis, we observe the general distribution of user movement from the center as illustrated in

Figure 11(b). In Ours, the distribution is slightly more spread toward the right, meaning users tend to step further away from the center. Comparing the two histograms using Jensen–Shannon divergence yielded about 10% ($\sqrt{\text{JSD}} = 0.106$, 95% CI [0.101, 0.116]). Figure 11(d) quantifies spatial distribution by dividing the area in (a) into a 5×5 grid and plotting the number of grids by percentage of trajectory points in log scale. Ours shows a more balanced distribution across the mid-range intervals and less concentration at the center grid than the Baseline.

6.6.2 Statistical Results of Subjective Measures. We ran the Shapiro–Wilk test to data in all conditions to analyze the normality of the data gathered. For measures that passed the normality test, which are Interest, and Message Understanding, we ran a paired samples t-test. For measures that did not pass the normality test, which are MySpace, YourSpace, Spatial Presence, VRSQ, and TPI, we ran the nonparametric Wilcoxon signed-rank tests to compare conditions. Measure YourSpace, Interest, and TPI showed significance as illustrated in Figure 12.

- **MySpace and YourSpace.** Using a modified Spatial situation model questionnaire, we measure two effects: understanding of my physical space (MySpace) and other users' spaces (YourSpace). Variable MySpace did not show significance under the Wilcoxon signed-rank test ($p = 0.227$) while variable YourSpace showed significance between conditions ($W = 51.500$, $z = -3.851$, $p < 0.001$). The mean value of the Ours ($m = 3.985$, $sd = 0.817$) was higher than Baseline ($m = 3.045$, $sd = 1.182$).
- **Spatial Presence** did not show significance between conditions ($p = 0.317$).
- **VRSQ** did not show significance between conditions ($p = 0.325$).
- **Interest** showed significance between conditions. A paired-samples t-test indicated that scores were significantly higher for Ours ($m = 6.126$, $sd = 0.644$) than for Baseline ($m = 5.866$, $sd = 0.644$), with $t(32) = -2.398$, $p = 0.022$.
- **TPI** showed significance between conditions ($W = 84.500$, $z = -2.811$, $p = 0.005$) under Wilcoxon signed-rank test. The mean value of Ours ($m = 5.398$, $sd = 0.987$) was higher than Baseline ($m = 4.961$, $sd = 1.046$).
- **Message Understanding** did not show significance between conditions ($p = 0.157$).

6.6.3 Interview Results. At the end of the study, we performed a semi-structured interview to ask and analyze the reasoning behind participants' thoughts in-depth, especially related to the two conditions they experienced. We asked four questions. Here, we provide summary of the results which details would be dealt within the discussion.

Question 1: Please compare the two conditions and explain how each one felt or what kind of experience they provided. Please explain freely. Participants typically explained the difference between conditions within the task context and reported higher understanding and presence toward Ours. Most participants highlighted three key aspects. First regarding enhanced spatial understanding. “When other users are explaining, it is more direct since I can see their spaces”(P12). Second is about increased sense of

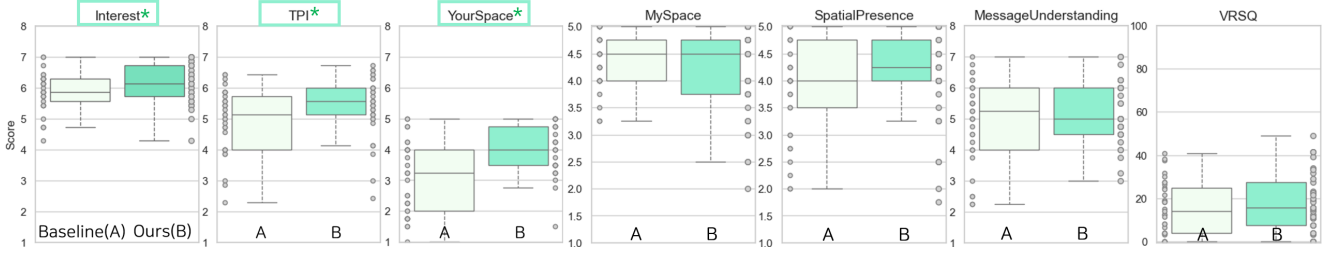


Fig. 12. Statistical results of comparisons between conditions for all measures. Factor Interest, TPI, and YourSpace showed significance as marked with an asterisk (*).

presence: “Condition Ours makes me feel like we are more together” (P29). Third is regarding a more immersive MR experience: “Baseline seems like augmented but condition Ours seems more mixed” (P26).

Question 2: Was there a more amusing condition? If so, which one and why? About four-fifths of participants (27 out of 33) reported Ours as more amusing. On the contrary, those who preferred Baseline over Ours generally cited their concentration levels, stating “Baseline was easier to concentrate” (P24), “Ours was initially interesting but I got tired more” (P14).

Question 3: Did you notice the visibility modulation feature of the system and how it affected your experience while performing the tasks? Participants understood the difference between conditions well, which affected their task performance. For Task 1 (introducing spaces), they generally preferred condition Ours. For Task 2 (block stacking), preferences varied. Participants who preferred Baseline mentioned that finding hidden blocks initially located in their own space but covered by other spaces was challenging. Those who preferred Ours noted that the blocks belonging to other spaces were not visible unless they were inside the shared space, which posed difficulties.

Question 4: If there were any features you thought would be fun to include in the system or any spatial experiences you believed would enhance it, please tell. Some participants suggested adding the ability to lock space visibility at specific moments during transitions, in order to prevent unintended changes caused by movement. As participants noted, “I want to choose only one room depending on who are introducing” (P4) or “Space keeps on changing... but I want to see specific parts of the space” (P6).

6.7 Discussion

Our study offers insights into the role of showing remote spaces and dynamic visibility modulation in enhancing user understanding and presence within multi-user telepresence environments.

6.7.1 Our System Enhanced Users’ Understanding of Both the Remote Spaces and Other Users’ Activities by Providing Spatial Context. Our findings indicate that showing remote spaces and dynamic visibility modulation significantly improved participants’ spatial awareness and understanding of remote environments. Participants repeatedly noted their impression of spaces changing through movement, with the visibility adapting in real-time. Several participants highlighted how the changing visibility based on movement supported their spatial understanding. “As the person explaining moved around and the room environment changed, I

could see how it differed from what I had imagined” (P5), “The room changes when you get closer, which made explanations feel much more realistic” (P21). Statistical results of the measure “MySpace” did not show significance between conditions. We can construe that the augmentation of the other remote spaces did not significantly harm the understanding of the user’s local space. However, the measure “YourSpace” showed significance aligning with participants’ qualitative remarks, showing a significant difference between conditions and demonstrating that our system enhances the understanding of other spaces. From these results, we support H1.

Message understanding did not show significance. As so, we reject H2. We conjecture that despite the absence of avatars in Baseline, participants could still engage in verbal communication through voice transfer and did not encounter much difficulty in understanding each other. Although the key message for the task itself was continuously transferred through voice, participants seemed to acquire extra awareness of others’ activities through visibility modulation. P11 directly stated that “It was good for figuring out what others were doing,” while P16 noted that “There was more in my field of view, so I could understand the situation visually as well as through explanations.” These observations suggest that users gained a contextual interpretation of an activity, which appears to affect a co-presence. P12 reported that in condition Ours, “it was easy to track other people’s movements, and I felt like we were together in the same space.” This extra information seems to affect participants to report higher social presence. With these qualitative remarks and the measure “TPI” showing statistical significance, we support H3.

6.7.2 Mixing Remote Spaces Provides Similar Spatial Presence as Only Observing the Local Space. While the p-value of the measure “Spatial Presence” was insignificant ($p = 0.317$), the absolute score of spatial presence revealed that participants reported a notable sense of spatial presence in both cases. Under a 5-point Likert scale ranging from 1 (“I do not agree at all”) to 5 (“I fully agree”), both conditions scored around 4 with $m = 3.886$, $sd = 0.914$ for Baseline and $m = 4.068$, $sd = 0.873$ for Ours. The visibility of other remote spaces might contribute to the overall sense of spatial presence, rather than diminishing it as might be expected with synthesized environments. During the interview, participants frequently (17 out of 33) mentioned how observing other spaces positively influenced their sense of spatial presence, especially when comparing Baseline and Ours. Despite the part of space being synthesized

through augmented remote spaces, participants reported similar levels of presence, indicating that synthesized spaces, even if it is dynamically changing, can provide equivalent spatial presence to real environments. Therefore, we support H4.

6.7.3 Our System Helped the Spaces Look More Interconnected, which Enhanced Users' Interests. We assumed that a significant portion of participants would have little previous experience with MR. Given that experiencing the MR environment itself would be new to the participants, we hypothesized that both conditions would be equally amusing in H5. However, users reported that Ours was more amusing than Baseline, supported by the statistical significance of the measure "Interest". We were able to induce some reasons from participants' comments from the interview. The fact that rooms are continuously connected to different configurations allows unusual experiences to participants and is approached as "fascinating" (P5), "more connected" (P10), "crossing between spaces" (P22), and "opening up like a curtain" (P25). These adjectives are specifically used within the line of visibility modulation. Studies have pointed out how interactivity can enhance users' engagement and excitement [Heeter 1992]. We believe this interactivity level can also work in the users' movement context, which changes the spatial visibility.

6.7.4 Our System did not Cause much Discomfort. For H6, we sought to determine whether our system causes unwanted disorientation or nausea. The VRSQ measure did not show significant differences between the two conditions, with scores of less than 20 for both conditions. From this observation, we argue that our system does not cause significant dizziness.

6.7.5 Under Our System, Users Make More Use of Their Own Space. In H7, we aimed to observe whether our system expanded the usage of spaces outside the shared space. We found that our system significantly increased both the average and maximum distances from the centroid, leading to larger space usage.

6.7.6 Color Difference. The color scheme was used not only for design reasons but also to help participants distinguish between different spaces. Several participants (P7, P14, P15) explicitly noted that the color difference supported room differentiation, suggesting that visual theming can serve as a low-effort spatial cue in MR environments. However, color transitions also introduced a few discomforts: Two participants reported slight dizziness due to the visual blending of colors across rooms. We conducted a between-subjects comparison for each condition's measure to see whether there is statistical significance between color groups. A statistical analysis ($n = 11$ per group) showed no significant impact of color on user experience, though the small sample size limits the strength of this conclusion. From this result, color did not appear to have a strong influence on spatial construction. However, future work may need to account for visual transitions when designing spatially mixed MR environments.

7 Limitations, Future Works, and Conclusion

Our system has several limitations to be addressed in future work. The primary one relates to the adjoined area. In the current system, users' spaces are sharply split, leading to divided furniture. The

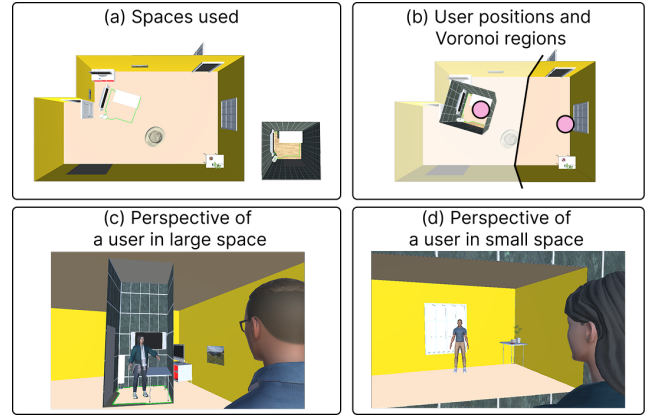


Fig. 13. Overlapping two rooms with extremely different sizes. (a) Top view of the rooms with shared space marked with a green line. (b) Conceptual Voronoi partitioning with user positions in pink color. The Voronoi split occurs inside the large room, disjoining the two rooms. (c) and (d) show the visible area from each local user's perspective.

split furniture might look cluttered from time to time. This limitation prompts a future research direction: exploring how to link split regions. Possible solutions would include temporarily hiding each space's furniture during transitions using diminished reality, blending spaces linearly as in SceneFusion [Wang et al. 2023], or employing generative methods like SpaceBlender [Numan et al. 2024]. We anticipate that these smooth transitions would enhance visual appeal, as reported in Numan et al. [2024].

We also present an extreme case in Figure 13, illustrating a scenario where two spaces differ significantly in size and the users are positioned far apart. In our method, the Voronoi split (b) introduces a spatial gap between the two environments. As a result, the user in the smaller space perceives the user in the larger space as if they are situated far in a disjointed room (d). This artifact could be mitigated by introducing an additional mechanism to prevent the spatial split from producing a gap. In cases where rooms have a large ceiling height difference, the local user's ceiling is partially exposed, as shown in Figure 14. In our setup, the local room is rendered as see-through, with the remote room visually overlaid. Consequently, a user in a lower-ceiling room can directly perceive the portion of the higher-ceiling room extending above their own ceiling, while a user in a higher-ceiling room can also perceive the ceiling of the lower room. Investigating the perceptual impact of such spatial gaps, including ceiling differences, and developing strategies to mitigate any potential negative effects remain an interesting avenue for future research.

A temporally fixed visibility mode has potential utility and is worth exploring in future work, particularly to better understand when static visibility is more beneficial than dynamic visibility. Insights can be drawn from participant feedback: five participants specifically expressed a desire for a fixed visibility mode to improve system usability. They noted that such a mode would be especially useful when the owner of a space is explaining a specific area, as temporarily fixing visibility can help maintain focus. This observation aligns with prior studies showing that even simple

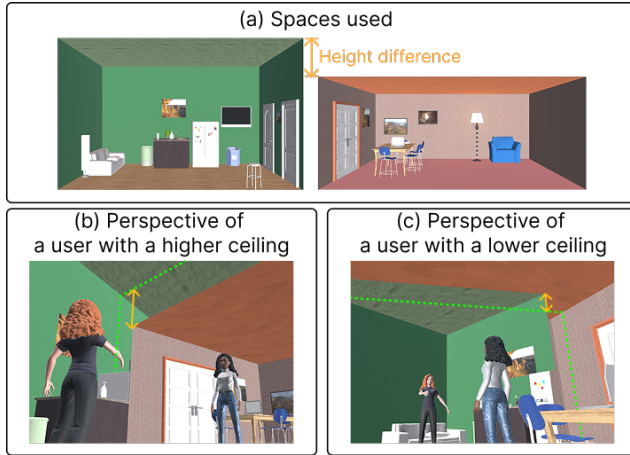


Fig. 14. Overlapping two rooms with large height gaps. (a) Side views. (b-c) Views from the room with a higher and a lower ceiling, respectively. The green line indicates the original Voronoi boundary, while the yellow arrow highlights the vertical ceiling height gap that makes the remote room visible above or below the local physical ceiling.

visualizations of another space can aid in understanding teaching contexts [Thoravi Kumaravel et al. 2019] and enhancing spatial presence [Numan et al. 2024].

Our user experiment was conducted with three participating rooms, but the impact of a significantly larger number of rooms with severe spatial clutter, as illustrated in Figure 6, remains to be explored. When many users move concurrently, dynamic visibility modulation may introduce substantial visual fatigue, which in turn can affect user behavior. Therefore, a valuable direction for future research is to systematically quantify the relationship between switch rate, perceived comfort, and factors such as group size or spatial layout.

Another limitation is the insufficient semantic understanding of each space. The current method optimizes using only empty spaces, without deeply comprehending the usage of furniture. Following Keshavarzi et al.'s [2022] suggestion, we could adopt semantic tagging of objects to create more functional spaces. Our system might generate overly complex spaces if too many users—say, 10—join. With such a large group, a more elaborate approach to finding shared spaces, such as defining multiple distributed shared areas [Kim et al. 2024a], would be possible.

Our implementation uses a pre-scanned, static 3D mesh of the environment, assuming that the objects within the space remain unchanged. However, with the advancement of scanning and reconstruction techniques, real-time scene reconstruction and networked updates could enable our approach to accommodate dynamic object changes.

In this work, we propose a new multi-user mixed reality telepresence system that modulates the visibility of multiple aligned spaces based on users' positions. Our system allows users to feel as if their physical rooms are interconnected, with each user's space attached to their position as they move. We found that our system increases the usable area within each physical space compared to existing methods. Participants in our user experiment reported

heightened amusement, spatial understanding, and spatial presence. We hope our system will pave the way for a more engaging multi-user telepresence environment that enhances spatial usability for each user.

References

- Guillaume Bataille, Abdelhadi Lammini, and Jean-Rémy Chardonnet. 2023. ARPuzzle: Evaluating the effectiveness of collaborative augmented reality. In *Proceedings of the 2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 642–651. DOI: <https://doi.org/10.1109/ISMAR59233.2023.00079>
- Stephan Beck, André Kunert, Alexander Kulik, and Bernd Froehlich. 2013. Immersive group-to-group telepresence. *IEEE Transactions on Visualization and Computer Graphics* 19, 4 (2013), 616–625. DOI: <https://doi.org/10.1109/TVCG.2013.33>
- Soojin Choi, Seokpyo Hong, Kyungmin Cho, Chaelin Kim, and Junyong Noh. 2023. Online avatar motion adaptation to morphologically-similar spaces. *Computer Graphics Forum* 42, 2 (2023), 13–24. arXiv: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/cgf.14740> DOI: <https://doi.org/10.1111/cgf.14740>
- Ben J. Congdon, Tuanfeng Wang, and Anthony Steed. 2018. Merging environments for shared spaces in mixed reality. In *Proceedings of the 24th ACM Symposium on Virtual Reality Software and Technology (VRST '18)*. ACM, New York, NY, USA, Article 11, 8 pages. DOI: <https://doi.org/10.1145/3281505.3281544>
- William Jay Conover. 1999. *Practical Nonparametric Statistics*. John Wiley & Sons.
- Carolina Cruz-Neira, Daniel J. Sandin, and Thomas A. DeFanti. 1993. Surround-screen projection-based virtual reality: The design and implementation of the CAVE. In *Proceedings of the 20th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '93)*. ACM, New York, NY, USA, 135–142. DOI: <https://doi.org/10.1145/166117.166134>
- Matt Deitke, Eli VanderBilt, Alvaro Herrasti, Luca Weihs, Kiana Ehsani, Jordi Salvador, Winson Han, Eric Kolve, Aniruddha Kembhavi, and Roozbeh Mottaghi. 2022. ProcTHOR: Large-scale embodied AI using procedural generation. In *Advances in Neural Information Processing Systems*. S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh (Eds.), Vol. 35, Curran Associates, Inc., 5982–5994. Retrieved from https://proceedings.neurips.cc/paper_files/paper/2022/file/27c546ab1e4f1d7d638e6a8dfbad9a07-Paper-Conference.pdf
- Zhi-Chao Dong, Xiao-Ming Fu, Zeshi Yang, and Ligang Liu. 2019. Redirected smooth mappings for multiuser real walking in virtual reality. *ACM Transactions on Graphics* 38, 5, Article 149 (Oct. 2019), 17 pages. DOI: <https://doi.org/10.1145/3345554>
- Terry Duncan Edward McAuley and Vance V. Tammen. 1989. Psychometric properties of the intrinsic motivation inventory in a competitive sport setting: A confirmatory factor analysis. *Research Quarterly for Exercise and Sport* 60, 1 (1989), 48–58. arXiv: <https://doi.org/10.1080/02701367.1989.10607413> DOI: <https://doi.org/10.1080/02701367.1989.10607413>
- Andrew J. Elliot and Markus A. Maier. 2014. Color psychology: Effects of perceiving color on psychological functioning in humans. *Annual Review of Psychology* 65, 1 (2014), 95–120.
- Daniel Immanuel Fink, Johannes Zagermann, Harald Reiterer, and Hans-Christian Jetter. 2022. Re-locations: Augmenting personal and shared workspaces to support remote collaboration in incongruent spaces. *Proceedings of the ACM on Human-Computer Interaction* 6, ISS, Article 556 (Nov. 2022), 30 pages. DOI: <https://doi.org/10.1145/3567709>
- Christian Frueh, Avneesh Sud, and Vivek Kwatra. 2017. Headset removal for virtual and mixed reality. In *ACM SIGGRAPH 2017 Talks (SIGGRAPH '17)*. ACM, New York, NY, USA, Article 80, 2 pages. DOI: <https://doi.org/10.1145/3084363.3085083>
- Simon Gibbs, Constantin Arapis, and Christian Breiteneder. 1999. TELEPORT - Towards immersive copresence. *Multimedia Systems* 7, 3 (05 1999), 214–221. DOI: <https://doi.org/10.1007/s005300050123>
- Jens Emil Sloth Grønbaek, Ken Pfeuffer, Eduardo Velloso, Morten Astrup, Melanie Isabel Sønderkær Pedersen, Martin Kjær, Germán Leiva, and Hans Gellersen. 2023. Partially blended realities: Aligning dissimilar spaces for distributed mixed reality meetings. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. ACM, New York, NY, USA, Article 456, 16 pages. DOI: <https://doi.org/10.1145/3544548.3581515>
- Chad Harms and Frank Biocca. 2004. Internal consistency and reliability of the networked mindsmeasure of social presence. Retrieved 14 Oct. 2024 from <https://api.semanticscholar.org/CorpusID:142957981>
- Carrie Heeter. 1992. Being there: The subjective experience of presence. *Presence: Teleoperators and Virtual Environments* 1, 2 (05 1992), 262–271. arXiv: <https://direct.mit.edu/pvar/article-pdf/1/2/262/1622433/pres.1992.1.2.262.pdf> DOI: <https://doi.org/10.1162/pres.1992.1.2.262>
- Andrew Irlitti, Mesut Latifoglu, Qiushi Zhou, Martin N. Reinoso, Thuong Hoang, Eduardo Velloso, and Frank Vetere. 2023. Volumetric mixed reality telepresence for real-time cross modality collaboration. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. ACM, New York, NY, USA, Article 101, 14 pages. DOI: <https://doi.org/10.1145/3544548.3581277>

- Sang-Bin Jeon, Soon-Uk Kwon, June-Young Hwang, Yong-Hun Cho, Hayeon Kim, Jinhung Park, and In-Kwon Lee. 2022. Dynamic optimal space partitioning for redirected walking in multi-user environment. *ACM Transactions on Graphics* 41, 4, Article 90 (July 2022), 14 pages. DOI: <https://doi.org/10.1145/3528223.3530113>
- Ying Jiang, Chang Yu, Tianyi Xie, Xuan Li, Yutao Feng, Huamin Wang, Minchen Li, Henry Lau, Feng Gao, Yin Yang, et al. 2024. VR-GS: A physical dynamics-aware interactive Gaussian splatting system in virtual reality. In *ACM SIGGRAPH 2024 Conference Papers (SIGGRAPH '24)*. ACM, New York, NY, USA, Article 78, 1 pages. DOI: <https://doi.org/10.1145/3641519.3657448>
- Dongsik Jo, Ki-Hong Kim, and Gerard Kim. 2015. SpaceTime: Adaptive control of the teleported avatar for improved AR tele-conference experience. *Computer Animation and Virtual Worlds* 26, 3–4 (04 2015), 259–269. DOI: <https://doi.org/10.1002/cav.1645>
- Brett Jones, Rajinder Sodhi, Michael Murdock, Ravish Mehra, Hrvoje Benko, Andrew Wilson, Eyal Ofek, Blair MacIntyre, Nikunj Raghuvanshi, and Lior Shapira. 2014. RoomAlive: Magical experiences enabled by scalable, adaptive projector-camera units. In *Proceedings of the 27th Annual ACM Symposium on User Interface Software and Technology (UIST '14)*. ACM, New York, NY, USA, 637–644. DOI: <https://doi.org/10.1145/2642918.2647383>
- Jiho Kang, Dongseok Yang, Taehee Kim, Yewon Lee, and Sung-Hee Lee. 2023. Real-time retargeting of deictic motion to virtual avatars for augmented reality telepresence. In *Proceedings of the 2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 885–893. DOI: <https://doi.org/10.1109/ISMAR59233.2023.00104>
- Naz Kaya and Helen H. Epps. 2004. Relationship between color and emotion: A study of college students. *College Student Journal* 38, 3 (2004), 396–405.
- M. Keshavarzi, A. Y. Yang, W. Ko, and L. Caldas. 2020. Optimization and manipulation of contextual mutual spaces for multi-user virtual and augmented reality interaction. In *Proceedings of the 2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. IEEE Computer Society, Los Alamitos, CA, USA, 353–362. DOI: <https://doi.org/10.1109/VR46266.2020.00055>
- Mohammad Keshavarzi, Michael Zollhoefer, Allen Y. Yang, Patrick Peluse, and Luisa Caldas. 2022. Mutual scene synthesis for mixed reality telepresence. arXiv:2204.00161 [cs.HC]. Retrieved from <https://arxiv.org/abs/2204.00161>
- Dooyoung Kim, Seonji Kim, Selin Choi, and Woontack Woo. 2024a. Spatial affordance-aware interactable subspace allocation for mixed reality telepresence. In *Proceedings of the 2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 1256–1265. DOI: <https://doi.org/10.1109/ISMAR62088.2024.00142>
- D. Kim and W. Woo. 2023. Edge-centric space rescaling with redirected walking for dissimilar physical-virtual space registration. In *Proceedings of the 2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE Computer Society, Los Alamitos, CA, USA, 829–838. DOI: <https://doi.org/10.1109/ISMAR59233.2023.00098>
- Hyun K. Kim, Jaehyun Park, Yeongcheol Choi, and Mungyeong Choe. 2018. Virtual reality sickness questionnaire (VRSQ): Motion sickness measurement index in a virtual reality environment. *Applied Ergonomics* 69 (2018), 66–73. DOI: <https://doi.org/10.1016/j.apergo.2017.12.016>
- Seonji Kim, Dooyoung Kim, Jae-Eun Shin, and Woontack Woo. 2024b. Object cluster registration of dissimilar rooms using geometric spatial affordance graph to generate shared virtual spaces. In *Proceedings of the 2024 IEEE Conference Virtual Reality and 3D User Interfaces (VR)*. 796–805. DOI: <https://doi.org/10.1109/VR58804.2024.00099>
- Yeonjun Kim, Hangil Park, Seungbae Bang, and Sung-Hee Lee. 2016. Retargeting human-object interaction to virtual avatars. *IEEE Transactions on Visualization and Computer Graphics* 22, 11 (2016), 2405–2412. DOI: <https://doi.org/10.1109/TVCG.2016.2593780>
- LaanConsultingCorp. 2024. 3D Scanner App. Retrieved 14 Oct. 2024 from <https://3dscannerapp.com/>
- Eike Langbehn, Frank Steinicke, Markus Lappe, Gregory F. Welch, and Gerd Bruder. 2018. In the blink of an eye: Leveraging blink-induced suppression for imperceptible position and orientation redirection in virtual reality. *ACM Transactions on Graphics* 37, 4, Article 66 (July 2018), 11 pages. DOI: <https://doi.org/10.1145/3197517.3201335>
- Jason Lawrence, Danb Goldman, Supreeth Achar, Gregory Major Blascovich, Joseph G. Desloge, Tommy Fortes, Eric M. Gomez, Sascha Häberling, Hugues Hoppe, Andy Huibers, et al. 2021. Project starline: A high-fidelity telepresence system. *ACM Transactions on Graphics* 40, 6, Article 242 (Dec. 2021), 16 pages. DOI: <https://doi.org/10.1145/3478513.3480490>
- Jason Lawrence, Ye Pan, Dan B. Goldman, Rachel McDonnell, Julie Robillard, Carol O'Sullivan, Yaser Sheikh, Michael Zollhoefer, and Jason Saragih. 2022. State of the art in telepresence. In *ACM SIGGRAPH 2022 Courses (SIGGRAPH '22)*. ACM, New York, NY, USA, Article 14, 159 pages. DOI: <https://doi.org/10.1145/3532720.3535673>
- Nicolas H. Lehment, Daniel Merget, and Gerhard Rigoll. 2014. Creating automatically aligned consensus realities for AR videoconferencing. In *Proceedings of the 2014 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 201–206. DOI: <https://doi.org/10.1109/ISMAR.2014.6948428>
- Matthew Lombard, Theresa B. Ditton, and Lisa Weinstein. 2009. Measuring presence: The temple presence inventory. In *Proceedings of the 12th Annual International Workshop on Presence*. International Society for Presence Research Los Angeles, CA, 1–15.
- LumaAI. 2024. LumaAI - Interactive Scenes. Retrieved 14 Oct. 2024 from <https://lumalabs.ai/interactive-scenes>
- Andrew Maimone and Henry Fuchs. 2011a. Encumbrance-free telepresence system with real-time 3D capture and display using commodity depth cameras. In *Proceedings of the 2011 10th IEEE International Symposium on Mixed and Augmented Reality*. 137–146. DOI: <https://doi.org/10.1109/ISMAR.2011.6092379>
- Andrew Maimone and Henry Fuchs. 2011b. A first look at a telepresence system with room-sized real-time 3d capture and life-sized tracked display wall. Retrieved 14 Oct. 2024 from <https://api.semanticscholar.org/CorpusID:30511926>
- Andrew Maimone, Xubo Yang, Nate Dierk, Andrei State, Mingsong Dou, and Henry Fuchs. 2013. General-purpose telepresence with head-worn optical see-through displays and projector-based lighting. In *2013 IEEE Virtual Reality (VR)*. 23–26. DOI: <https://doi.org/10.1109/VR.2013.6549352>
- Hideobu Matsuki, Riku Murai, Paul H. J. Kelly, and Andrew J. Davison. 2024. Gaussian splatting SLAM. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 18039–18048.
- Nels Numan, Shwetha Rajaram, Balasaravanan Thoravi Kumaravel, Nicolai Marquardt, and Andrew D. Wilson. 2024. SpaceBlender: Creating context-rich collaborative spaces through generative 3D scene blending. *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology UIST '24*. 1–25.
- Sergio Orts-Escobedo, Christoph Rhemann, Sean Fanello, Wayne Chang, Adarsh Kowdle, Yuri Degtyarev, David Kim, Philip L. Davidson, Sameh Khamis, Mingsong Dou, et al. 2016. Holoportation: Virtual 3D teleportation in real-time. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology (UIST '16)*. ACM, New York, NY, USA, 741–754. DOI: <https://doi.org/10.1145/2984511.2984517>
- Tomislav Pejisa, Julian Kantor, Hrvoje Benko, Eyal Ofek, and Andrew Wilson. 2016. Room2Room: Enabling life-size telepresence in a projected augmented reality environment. In *Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing (CSCW '16)*. ACM, New York, NY, USA, 1716–1725. DOI: <https://doi.org/10.1145/2818048.2819965>
- Thammathip Piumsomboon, Gun A. Lee, Jonathan D. Hart, Barrett Ens, Robert W. Lindeman, Bruce H. Thomas, and Mark Billinghurst. 2018. Mini-Me: An adaptive avatar for mixed reality remote collaboration. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI'18)*. ACM, New York, NY, USA, 1–13. DOI: <https://doi.org/10.1145/3173574.3173620>
- Polycam. 2024. Polycam - LiDAR & 3D Scanner for iPhone & Android. Retrieved 14 Oct. 2024 from <https://poly.cam/>
- Ramesh Raskar, Greg Welch, Matt Cutts, Adam Lake, Lev Stesin, and Henry Fuchs. 1998. The office of the future: A unified approach to image-based modeling and spatially immersive displays. In *Proceedings of the 25th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '98)*. ACM, New York, NY, USA, 179–188. DOI: <https://doi.org/10.1145/280814.280861>
- Sharif Razaque, Zachariah Kohn, and Mary C. Whittton. 2001. Redirected walking. In *Eurographics 2001 - Short Presentations*. Eurographics Association. DOI: <https://doi.org/10.2312/egs.20011036>
- S. S. Shapiro and M. B. Wilk. 1965. An analysis of variance test for normality (complete samples)†. *Biometrika* 52, 3–4 (12 1965), 591–611. arXiv:<https://academic.oup.com/biomet/article-pdf/52/3-4/591/962907/52-3-4-591.pdf> DOI: <https://doi.org/10.1093/biomet/52.3-4.591>
- Agnieszka Sorokowska, Piotr Sorokowski, Peter Hilpert, Katarzyna Cantarero, Tomasz Frackowiak, Ahmad Alghraibeh, Richmond Aryeetey, Anna Bertoni, Karim Betache, Sheyla Blumen, et al. 2017. Preferred interpersonal distances: A global comparison. *Journal of Cross-Cultural Psychology* 48, 4 (03 2017), 002202211769803. DOI: <https://doi.org/10.1177/0022022117698039>
- Misha Sra, Aske Mottelson, and Pattie Maes. 2018. Your place and mine: Designing a shared VR experience for remotely located users. In *Proceedings of the 2018 Designing Interactive Systems Conference (DIS '18)*. ACM, New York, NY, USA, 85–97. DOI: <https://doi.org/10.1145/3196709.3196788>
- Qi Sun, Anjul Patney, Li-Yi Wei, Omer Shapira, Jingwan Lu, Paul Asente, Suwen Zhu, Morgan Mcguire, David Luebke, and Arie Kaufman. 2018. Towards virtual reality infinite walking: dynamic saccadic redirection. *ACM Transactions on Graphics* 37, 4, Article 67 (July 2018), 13 pages. DOI: <https://doi.org/10.1145/3197517.3201294>
- Balasaravanan Thoravi Kumaravel, Fraser Anderson, George Fitzmaurice, Bjørn Hartmann, and Tovi Grossman. 2019. Loki: Facilitating remote instruction of physical tasks using bi-directional mixed-reality telepresence. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology (UIST '19)*. ACM, New York, NY, USA, 161–174. DOI: <https://doi.org/10.1145/3332165.3347872>
- Huayuan Tian, Gun A. Lee, Huidong Bai, and Mark Billinghurst. 2023. Using virtual replicas to improve mixed reality remote collaboration. *IEEE Transactions on Visualization and Computer Graphics* 29, 5 (2023), 2785–2795. DOI: <https://doi.org/10.1109/TVCG.2023.3247113>
- Peter Vorderer, Werner Wirth, Feliz Gouveia, Frank Biocca, Timo Saari, Lutz Jäncke, Saskia Böcking, Holger Schramm, Andre Gysbers, Tilo Hartmann, et al. 2004. MEC spatial presence questionnaire (MEC-SPQ, English and German version): Short documentation and instructions for application. DOI: <https://doi.org/10.13140/RG.2.2.26232.42249>. Accessed 14 Oct. 2024.

- Miao Wang, Yi-Jun Li, Jinchuan Shi, and Frank Steinicke. 2023. SceneFusion: Room-scale environmental fusion for efficient traveling between separate virtual environments. *IEEE Transactions on Visualization and Computer Graphics* 30, 8 (2023), 4615–4630. DOI: <https://doi.org/10.1109/TVCG.2023.3271709>
- Peng Wang, Xiaoliang Bai, Mark Billingham, Shusheng Zhang, Xiangyu Zhang, Shuxia Wang, Weiping He, Yuxiang Yan, and Hongyu Ji. 2021. AR/MR remote collaboration on physical tasks: A review. *Robotics and Computer-Integrated Manufacturing* 72, C (2021), 102071. DOI: <https://doi.org/10.1016/j.rcim.2020.102071>
- Xuanyu Wang, Hui Ye, Christian Sandor, Weizhan Zhang, and Hongbo Fu. 2022. Predict-and-drive: Avatar motion adaption in room-scale augmented reality telepresence with heterogeneous spaces. *IEEE Transactions on Visualization and Computer Graphics* 28, 11 (2022), 3705–3714. DOI: <https://doi.org/10.1109/TVCG.2022.3203109>
- Frank Wilcoxon. 1945. Individual comparisons by ranking methods. *Biometrics Bulletin* 1, 6 (1945), 80–83. Retrieved from <http://www.jstor.org/stable/3001968>
- Linning Xu, Vasu Agrawal, William Laney, Tony Garcia, Aayush Bansal, Changil Kim, Samuel Rota Bulò, Lorenzo Porzi, Peter Kotschieder, Aljaž Božič, et al. 2023. VR-NeRF: High-fidelity virtualized walkable spaces. In *SIGGRAPH Asia 2023 Conference Papers (SA '23)*. ACM, New York, NY, USA, Article 43, 12 pages. DOI: <https://doi.org/10.1145/3610548.3618139>
- Dongseok Yang, Jiho Kang, Taehei Kim, and Sung-Hee Lee. 2024. Visual guidance for user placement in avatar-mediated telepresence between dissimilar spaces. *IEEE Transactions on Visualization and Computer Graphics* 30, 12 (2024), 7558–7570. DOI: <https://doi.org/10.1109/tvcg.2024.3354256>
- Leonard Yoon, Dongseok Yang, Jaehyun Kim, Choongho Chung, and Sung-Hee Lee. 2020. Placement retargeting of virtual avatars to dissimilar indoor environments. *IEEE Transactions on Visualization and Computer Graphics* 28, 3 (08 2020), 1619–1633. DOI: <https://doi.org/10.1109/TVCG.2020.3018458>

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